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Elucidating the role of metal ions in carbonic anhydrase catalysis

Jin Kyun Kim¹, Cheol Lee¹, Seon Woo Lim¹, Aniruddha Adhikari¹, Jacob T. Andring², Robert McKenna², Cheol-Min Ghim¹ & Chae Un Kim¹✉

Why metalloenzymes often show dramatic changes in their catalytic activity when subjected to chemically similar but non-native metal substitutions is a long-standing puzzle. Here, we report on the catalytic roles of metal ions in a model metalloenzyme system, human carbonic anhydrase II (CA II). Through a comparative study on the intermediate states of the zinc-bound native CA II and non-native metal-substituted CA IIs, we demonstrate that the characteristic metal ion coordination geometries (tetrahedral for Zn^{2+} , tetrahedral to octahedral conversion for Co^{2+} , octahedral for Ni^{2+} , and trigonal bipyramidal for Cu^{2+}) directly modulate the catalytic efficacy. In addition, we reveal that the metal ions have a long-range (~ 10 Å) electrostatic effect on restructuring water network in the active site. Our study provides evidence that the metal ions in metalloenzymes have a crucial impact on the catalytic mechanism beyond their primary chemical properties.

¹Department of Physics, Ulsan National Institute of Science and Technology (UNIST), Ulsan 44919, Republic of Korea. ²Department of Biochemistry and Molecular Biology, College of Medicine, University of Florida, Gainesville, FL 32610, USA. ✉email: cukim@unist.ac.kr

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闡明金屬離子在碳酸酐酶催化中的作用

Jin Kyun Kim¹、Cheol Lee¹、Seon Woo Lim¹、Aniruddha Adhikari¹、Jacob T. Andring²、Robert McKenna²、Cheol-Min Ghim¹ 和 Chae Un Kim¹

為什麼金屬酵素在經歷化學上相似但非天然的金屬取代時，其催化活性經常表現出巨大的變化，這是一個長期存在的謎團。在這裡，我們報告了金屬離子在模型金屬酶系統——人類碳酸酐酶 II (CA II) 中的催化作用。透過鋅結合的天然CA II和非天然金屬取代的CA II的中間態的比較研究，我們證明了特徵金屬離子配位幾何形狀（Zn²⁺為四面體，Co²⁺為四面體到八面體轉換，Ni²⁺為八面體，Cu²⁺為三角錐體）直接調節催化效率。此外，我們發現金屬離子對活性位點的水網絡重組具有長程（~10 Å）靜電效應。我們的研究提供的證據表明，金屬酶中的金屬離子對催化機制的影響超出了其主要化學性質。

¹Department of Physics, Ulsan National Institute of Science and Technology (UNIST), Ulsan 44919, Republic of Korea. ²Department of Biochemistry and Molecular Biology, College of Medicine, University of Florida, Gainesville, FL 32610, USA. ✉email: cukim@unist.ac.kr

Metalloproteins are ubiquitous in nature and play indispensable roles in key biological processes, such as DNA synthesis, chemical signaling, and cellular metabolism^{1,2}. Due to their versatile chemical reactivity (acidity, electrophilicity, and/or nucleophilicity), incorporated metal ions add functionality to proteins and help catalyze some of the most intricate reactions in nature^{3,4}. The issues of metal binding affinity and specificity of metal ions to proteins have been studied based on the metal coordination stereochemistry^{5,6} and semi-empirical and qualitative theories such as hard and soft acids and bases principle of Parr and Pearson⁷ and Irving–Williams series of divalent ion stability^{8,9}. However, the role of metal ions in the functioning of proteins and the metal–protein relationships remain unclear at the atomic level. For example, metalloenzymes substituted by non-native metal ions often exhibit drastically different catalytic activities^{10,11}, even when the substituted metal ions show chemical features broadly similar to the native one, such as ionic charge/size/mass, redox potential, electronic configuration, and allowed coordination geometry.

Among the various types of metalloenzymes, carbonic anhydrase (CA), the first enzyme recognized to contain zinc, is ubiquitous across all kingdoms of life and one of the most catalytically efficient enzymes ever known^{12–16}. The enzyme catalyzes the reversible hydration of carbon dioxide (CO_2) and thereby plays a critical role in respiration, particularly in the CO_2 transport by way of blood-dissolved bicarbonate (HCO_3^-), and in intracellular pH homeostasis by maintaining $\text{CO}_2/\text{HCO}_3^-$ equilibrium. Within the wide classes of CA, CA II from human is well-suited to serve as a model system for investigating the role of metal ions because its overall structure is well-refined with atomic resolution ($\sim 1.0 \text{ \AA}$). It possesses a well-defined active site containing a single metal-binding site (Fig. 1a, b), and the kinetic rates and fine details of the enzymatic mechanism have been studied extensively^{17–21} (Fig. 1c).

The active site of CA II is located at the base cavity of a $15 \text{ \AA}$ depth from the surface and is further subdivided into three regions comprised of hydrophobic and hydrophilic regions, with an entrance conduit (EC) in-between^{22–25} (Fig. 1a). These regions are responsible for substrate binding, proton transfer, and substrate/product/water exchange during catalysis, respectively (Fig. 1b). The active site zinc ion is tetrahedrally coordinated to the protein by the imidazole groups of three histidine residues, with the remaining tetrahedral site occupied by a solvent molecule (water or hydroxide ion, depending upon pH). The catalytic zinc ion in CA II serves as a Lewis acid; its primary role is to lower the pK_a of the Zn-bound water from 10 to 7, allowing the formation of a zinc-bound hydroxide ion at physiological pH²⁶. The zinc ion can be substituted by other physiologically relevant transition metal ions such as Co^{2+} , Ni^{2+} , Cu^{2+} , Cd^{2+} , and Mn^{2+} which results in drastic changes in the catalytic activity of CA II ($\sim 50\%$ active to completely inactive)²¹. It has been also reported that the metal substitutions may induce alternative catalytic activities of CA II other than $\text{CO}_2/\text{HCO}_3^-$ conversion²⁷, for instance, reduction of nitrite to nitric oxide in presence of copper²⁸.

Previous structural studies had suggested that different metal coordination geometries in the non-native CA II may play an important role in their catalysis^{29,30}, but no clear evidence was presented to support such a claim. Our present study focuses on investigating the detailed structural changes in CA II during the $\text{CO}_2/\text{HCO}_3^-$ conversion catalysis and correlating these variations to the relevant catalytic mechanisms. These experimental insights

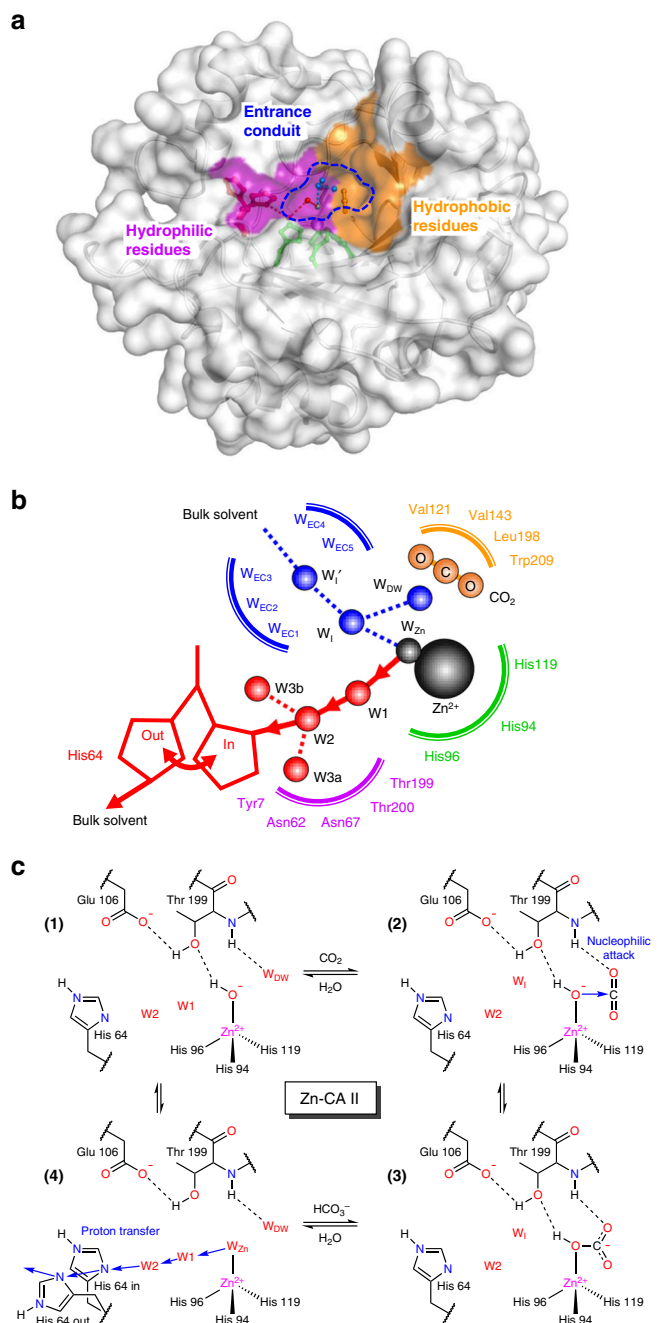


Fig. 1 Structure of native carbonic anhydrase II (Zn-CA II) and its catalytic mechanism. **a** The active site consists of zinc binding site, hydrophobic/hydrophilic regions, and entrance conduit (EC). **b** The water networks in the active site are responsible for the proton transfer (red) and substrate/product/water exchange (blue) during enzyme catalysis. **c** The CO_2 hydration reaction mechanism of Zn-CA II. First, CO_2 binds to the active site, leading to a nucleophilic attack by the zinc-bound hydroxyl ion onto CO_2 . HCO_3^- thus formed is subsequently displaced by the water molecule inflowing through EC. The HCO_3^- molecule likely binds to Zn^{2+} ion in a monodentate mode and its OH group is held at the Zn^{2+} ion due to the hydrogen bonding with Thr199^{52, 53}. This product binding configuration leads to a weak interaction between the product and Zn^{2+} ion, thereby facilitating fast product dissociation⁵⁴. Finally, proton transfer occurs via the network ($\text{W}_{\text{Zn}} \rightarrow \text{W}_1 \rightarrow \text{W}_2 \rightarrow \text{His64}$) provided by the protein scaffold.

金屬蛋白在自然界中無所不在，在 DNA 合成、化學訊號傳導和細胞代謝等關鍵生物過程中發揮著不可或缺的作用^{1,2}。由於其多功能的化學反應性（酸性、親電性和/或親核性），摻入的金屬離子可以為蛋白質添加功能，並有助於催化自然界中一些最複雜的反應^{3,4}。基於金屬配位立體化學^{5,6}以及半經驗和定性理論，如 Parr 和 Pearson 的硬酸和軟酸和鹼原理⁷以及二價離子穩定性的 Irving-Williams 系列^{8,9}，研究了金屬結合親和力和金屬離子與蛋白質的特異性問題。然而，金屬離子在蛋白質功能中的作用以及金屬-蛋白質關係在原子層面上仍不清楚。例如，被非天然金屬離子取代的金屬酶通常表現出截然不同的催化活性^{10,11}，即使取代的金屬離子表現出與天然金屬離子大致相似的化學特徵，例如離子電荷/尺寸/質量、氧化還原電位、電子構型和允許的配位幾何形狀。

在各種類型的金屬酶中，碳酸酐酶（CA）是第一種被認為含有鋅的酶，在所有生命領域中普遍存在，並且是迄今為止已知的催化效率最高的酶之一¹²⁻¹⁶。此酵素催化二氧化碳（CO₂）的可逆水合，從而在呼吸中發揮關鍵作用，特別是在透過血液溶解的碳酸氫鹽（HCO₃⁻）運輸 CO₂ 中，以及透過維持 CO₂/HCO₃⁻ 平衡來維持細胞內 pH 穩態。在廣泛的 CA 類別中，來自人類的 CA II 非常適合作為研究金屬離子作用的模型系統，因為其整體結構經過原子分辨率（~1.0 Å）的精細化處理。它擁有一個明確的活性位點，其中包含單一金屬結合位點（圖1a, b），並且酶促機制的動力學速率和細節已被廣泛研究¹⁷⁻²¹（圖1c）。

CA II 的活性位點位於距表面 15 Å 深度的基腔處，並進一步細分為由疏水區和親水區組成的三個區域，入口導管（EC）位於 22-25 之間（圖1a）。這些區域分別負責催化過程中的底物結合、質子轉移和底物/產物/水交換（圖1b）。活性位點鋅離子透過三個組胺酸殘基的咪唑基團與蛋白質進行四面體配位，其餘四面體位點被溶劑分子（水或氫氧根離子，取決於 pH 值）佔據。CA II 中的催化鋅離子充當路易斯酸；它的主要作用是將鋅結合水的 pK_a 從 10 降低至 7，從而在生理 pH 26 下形成鋅結合的氫氧根離子。鋅離子可以被其他生理相關的過渡金屬離子取代，例如 Co²⁺、Ni²⁺、Cu²⁺、Cd²⁺ 和 Mn²⁺，這會導致 CA II 的催化活性發生巨大變化（~50% 活性到完全失活）²¹。另據報導，金屬取代可能會誘導 CA I 除 CO₂/HCO₃⁻ 轉化之外的其他催化活性²⁷，例如，在銅存在下將亞硝酸鹽還原為一氧化氮²⁸。

先前的結構研究表明，非天然 CA II 中不同的金屬配位幾何形狀可能在其催化作用中發揮重要作用^{29,30}，但沒有明確的證據支持這種說法。我們目前的研究重點是研究 CO₂/HCO₃⁻ 轉化催化過程中 CA II 的詳細結構變化，並將這些變化與相關催化機制相關聯。這些實驗見解

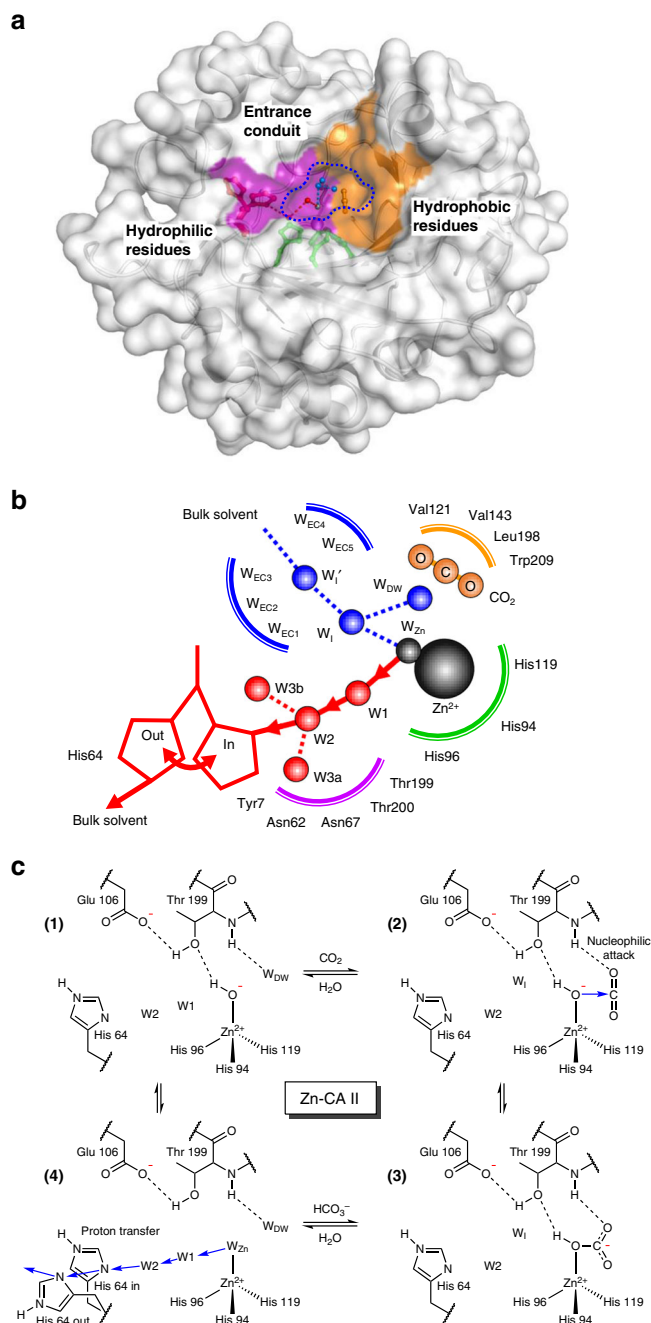


圖1 天然碳酸酐酶II (Zn-CA II) 的結構及其催化機制。a 活性位點由鋅結合位點、疏水/親水區域和入口導管 (EC) 組成。b 活性位點中的水網絡負責酵素催化過程中的質子轉移（紅色）和底物/產物/水交換（藍色）。c Zn-CA II 的 CO₂ 水合反應機制。首先，CO₂ 與活性位點結合，導致鋅結合的羥基離子對 CO₂ 進行親核攻擊。由此形成的 HCO₃⁻ 隨後被流經 EC 的水分子取代。HCO₃⁻ 分子可能以單齒模式與 Zn²⁺ 離子結合，並且由於與 Thr199 形成氫鍵，其 OH 基團保留在 Zn²⁺ 離子上，53。這種產物結合構型導致產物與 Zn²⁺ 離子之間的弱相互作用，進而促進產物快速解離 54。最後，質子轉移是透過蛋白質支架提供的網絡 (W_{Zn} → W₁ → W₂ → His64) 發生。

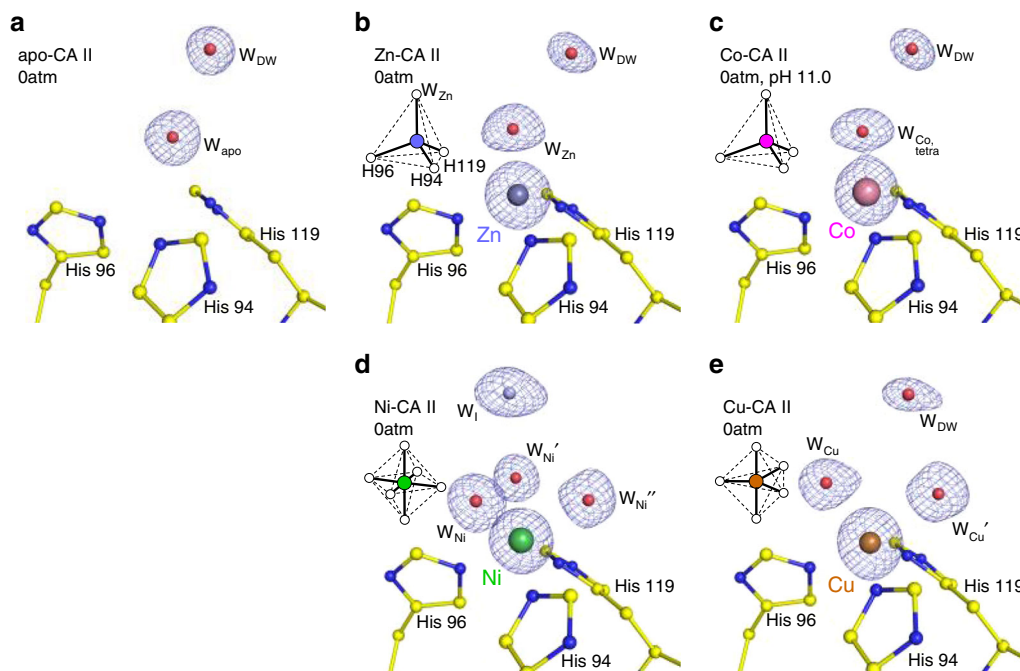


Fig. 2 Metal coordination geometry in CA II without CO₂ pressurization. **a** In apo-CA II, the metal binding site is vacant. **b, c** Zn- and Co-CA II show tetrahedral, **d** Ni-CA II octahedral, and **e** Cu-CA II trigonal bipyramidal coordination geometry. The electron density ($2F_o - F_c$, blue) is contoured at 2.2σ . All structures were obtained at pH 7.8 except for (**c**) which is obtained at pH 11.0. The intermediate water (W_i) in (**d**) is colored in steel blue for clarity.

offer us a fresh peek into the origin of the activity alterations caused by non-native metal substitutions.

To study the role of metal ions in CA II, we selected four divalent transition-metal ions (Zn^{2+} , Co^{2+} , Ni^{2+} , and Cu^{2+}) that induce drastic changes in CA II activity (100%, ~50%, ~2%, and 0%, respectively)^{31,32}. The catalytic intermediate states of the metal-free (apo, as a control) and the four metal-bound CA IIs were prepared by cryocooling protein crystals under CO₂ pressures from 0 (no CO₂ pressurization) to 20 atm^{33,34}.

We show that the characteristic metal ion coordination geometries directly modulate the catalytic processes, including substrate binding, its conversion to product, and product binding. In addition, we reveal that the metal ions have a long-range (~10 Å) electrostatic effect on restructuring the water network at the active site, affecting the product displacement and the proton transfer process. The cumulative effect of such alterations provides mechanistic insights into the overall reduction of the enzymatic activity in the non-native metal-substituted CA IIs.

Results

The role of metal ion coordination geometries. The coordination geometry around the metal binding site in CA II, when no CO₂ pressure is applied, is shown in Fig. 2. The metal-free apo-CA II shows an electron density map reflecting the presence of a water molecule in the metal binding site (Fig. 2a). In Zn-CA II and Co-CA II (pH 11.0), the metal ions display tetrahedral coordination with three histidine residues (His94, His96, and His119) and a water molecule (Fig. 2b, c). In contrast, Ni-CA II contains three bound water molecules, completing an octahedral (hexa-coordinate) geometry (Fig. 2d). Finally, Cu-CA II possesses two bound water molecules, arranged in trigonal bipyramidal (penta-coordinate) geometry (Fig. 2e).

Next, we investigated the effect of metal coordination geometry on the efficacy of substrate (CO₂) and product (HCO₃⁻) binding. The apo- and Zn-CA II structures cryocooled at 20 atm CO₂ pressure are shown in Fig. 3. The apo-CA II shows a clear binding

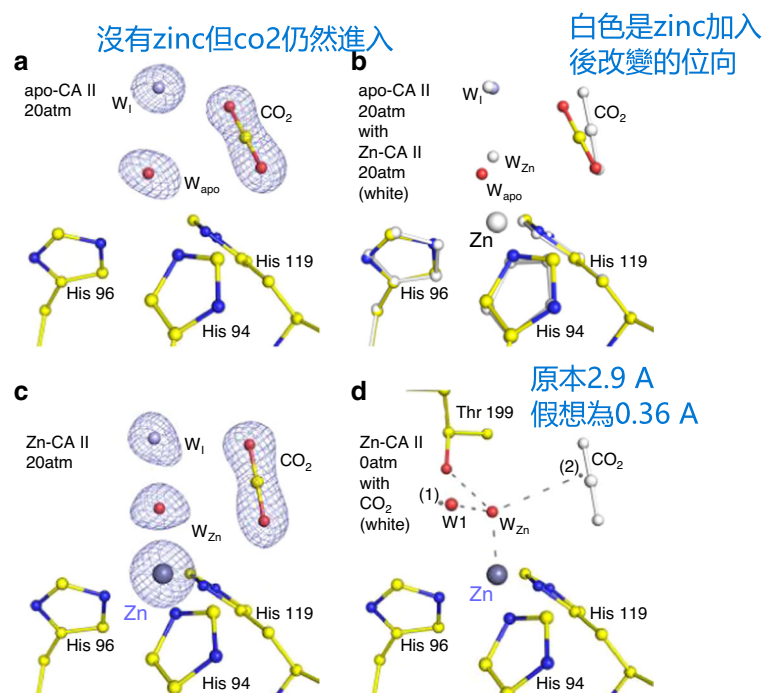


Fig. 3 Substrate/product binding in apo- and Zn-CA II. The intermediate water (W_i) is colored in steel blue for clarity. The electron density ($2F_o - F_c$, blue) is contoured at 2.2σ . **a, b** At 20 atm of CO₂ pressure, apo-CA II shows clear binding of CO₂ without the need of Zn^{2+} ion. **c** Zn-CA II shows similar binding of CO₂ as in apo-CA II while maintaining tetrahedral metal coordination. **d** Upon CO₂ binding (white) in Zn-CA II, W_{Zn} is located at the center of the hypothetical tetrahedral arrangement made up of Zn^{2+} ion, Thr199-Oγ1, position (1) (close to W_1), and position (2) (close to the carbon atom in CO₂). In this configuration, a hybridized lone pair in W_{Zn} directly faces CO₂ molecule at a distance, appropriate for efficient nucleophilic attack. Distance between the position (2) and C atom of CO₂ is merely 0.36 Å.

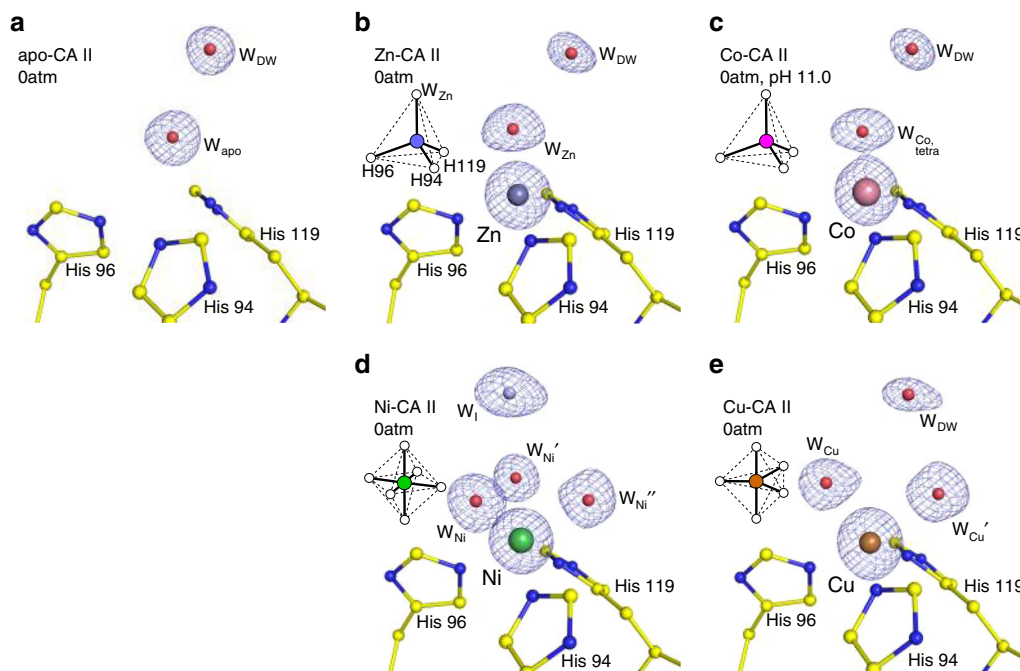


圖 2 沒有 CO_2 加壓的 CA II 中的金屬配位幾何形狀。a 在 apo-CA II 中，金屬結合位點是空的。b、c Zn-和 Co-CA II 顯示四面體、d Ni-CA II 八面體和 e Cu-CA II 三角雙錐配位幾何形狀。電子密度 ($2F_o - F_c$, 藍色) 的輪廓為 2.2σ 。所有結構均在 pH 7.8 下獲得，除了 (c) 是在 pH 11.0 下獲得的。為了清晰起見，(d) 中的中間水 (W_1) 被著色為鋼藍色。

讓我們對非天然金屬替代引起的活性改變的起源有一個新的了解。

為了研究金屬離子在 CA II 中的作用，我們選擇了四種二價過渡金屬離子 (Zn^{2+} 、 Co^{2+} 、 Ni^{2+} 和 Cu^{2+})，它們會引起 CA II 活性（分別為 100%、~50%、~2% 和 20%）的劇烈變化的劇烈變化。無金屬（apo，作為對照）和四種金屬結合的 CA II 的催化中間態是透過在 0（無 CO_2 加壓）至 20 atm^{33,34} 的 CO_2 壓力下冷凍蛋白質晶體來製備的。

我們表明，特徵性金屬離子配位幾何形狀直接調節催化過程，包括底物結合、其轉化為產物以及產物結合。此外，我們發現金屬離子對活性位點水網絡的重構具有長程（~10 Å）靜電效應，影響產物位移和質子轉移過程。這種改變的累積效應為非天然金屬取代的 CA II 中酶活性的總體降低提供了機制見解。

結果

金屬離子配位幾何形狀的作用。當不施加 CO_2 壓力時，CA I 中金屬結合位點周圍的配位幾何形狀如圖 2 所示。不含金屬的 apo-CA II 顯示了反映金屬結合位點中水分子存在的電子密度圖（圖 2a）。在 Zn-CA II 和 Co-CA II（pH 11.0）中，金屬離子與三個組胺酸殘基（His94、His96 和 His119）和水分子呈現四面體配位（圖 2b, c）。相較之下，Ni-CA II 包含三個結合水分子，形成八面體（六座標）幾何形狀（圖 2d）。最後，Cu-CA II 擁有兩個結合水分子，排列成三角雙錐（五配位）幾何形狀（圖 2e）。

接下來，我們研究了金屬配位幾何形狀對底物 (CO_2) 和產物 (HCO_3^-) 結合功效的影響。在 20 atm CO_2 壓力下冷凍冷卻的 apo- 和 Zn-CA II 結構如圖 3 所示。apo-CA II 顯示出清晰的結合

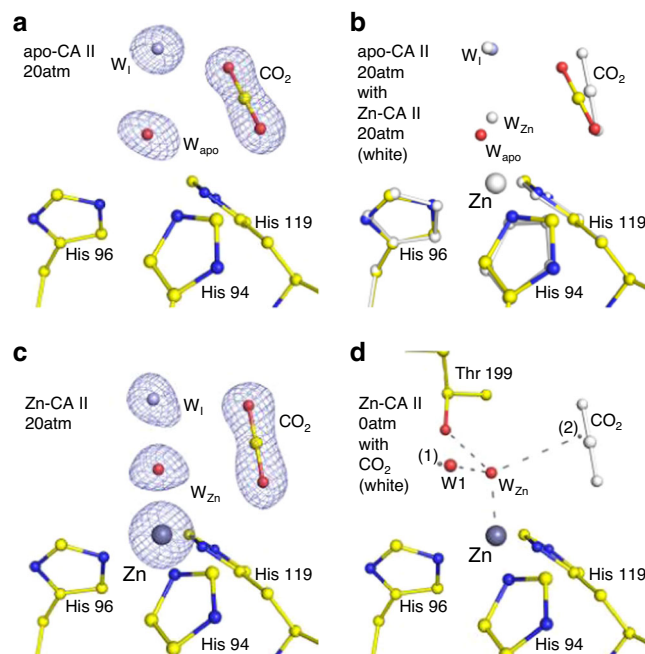


圖 3 apo- 和 Zn-CA II 中的底物/產物結合。為了清晰起見，中間水 (W_1) 的顏色為鋼藍色。電子密度 ($2F_o - F_c$, 藍色) 的輪廓為 2.2σ 。a, b 在 20 大氣壓的 CO_2 壓力下，apo-CA II 顯示與 CO_2 的清晰結合，而不需要 Zn^{2+} 離子。c Zn-CA II 顯示出與 apo-CA II 類似的 CO_2 結合，同時保持四面體金屬配位。d 在 Zn-CA II 中 CO_2 結合（白色）時， W_{Zn} 位於由 Zn^{2+} 離子、Thr199-O γ 1、位置 (1)（靠近 W_1 ）和位置 (2)（靠近 CO_2 中的碳原子）組成的假設四面體排列的中心。在這個配置中， W_{Zn} 中的雜化孤對電子直接面對 CO_2 分子，距離適合有效的親核攻擊。(2) 位元與 CO_2 的 C 原子之間的距離僅為 0.36 Å。

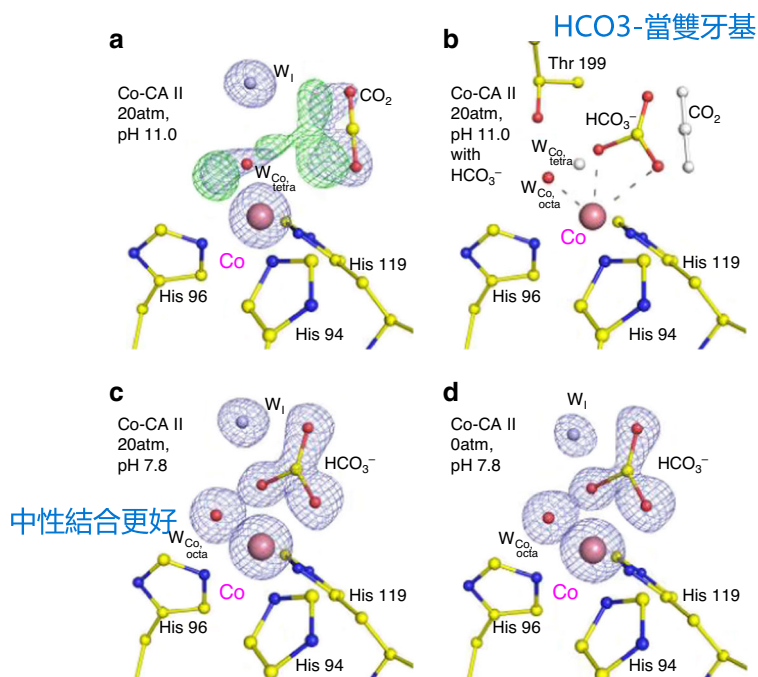


Fig. 4 Substrate/product binding in Co-CA II. The intermediate water (W_I) is colored in steel blue for clarity. The electron density ($2F_o - F_c$, blue) and the difference map ($F_o - F_c$, green) are contoured at 2.2σ and 7.0σ , respectively. **a, b** At 20 atm of CO_2 pressure, Co-CA II at pH 11.0 shows superposition of CO_2 binding (~50% occupancy, white) with tetrahedral coordination and HCO_3^- binding (~50% occupancy) with octahedral coordination. **c, d** Co-CA II at pH 7.8 shows complete binding of HCO_3^- , showing octahedral coordination even in absence of added CO_2 . It is likely that the captured HCO_3^- is converted from the CO_2 absorbed in the crystal from ambient air.

of CO₂ molecule, replacing deep water, W_{DW} present within the active site, suggesting that the CO₂ binding at the active site is mostly dictated by the metal-free protein scaffold (Fig. 3a). The native holoenzyme Zn-CA II shows CO₂ binding almost identical to that in apo-CA II (Fig. 3b, c). The CO₂ molecule is located 2.9 Å away from the Zn-bound water (W_{Zn}), in a configuration conducive for the nucleophilic attack (Fig. 3d).

In Co-CA II (pH 11.0) cryocooled at 20 atm CO₂ pressure, dual binding of CO₂ and HCO₃[−] is observed (Fig. 4a). Upon CO₂ binding, the tetrahedral coordination is maintained, but an unusual expansion to octahedral coordination is observed upon HCO₃[−] binding (Fig. 4b). In the transformed octahedral geometry, the HCO₃[−] molecule is bound in a bidentate mode to the Co²⁺ ion along with an additional water molecule. Compared to the monodentate binding mode in Zn-CA II, the negative charge on the bidentate HCO₃[−] can be distributed among the two oxygen atoms bound to Co²⁺ ion, allowing stronger product binding to the metal ion (Supplementary Fig. 1a–f). Unlike Zn-CA II (Supplementary Fig. 2), the Co-CA II intermediates obtained at different pH values (7.8 and 11.0) reveal that the HCO₃[−] molecule is firmly bound to Co²⁺ ion with full occupancy at lower pH (Fig. 4c, d), but this binding affinity weakens as pH increases (Fig. 4a). The result suggests that, during the catalytic cycle, deprotonation of the Co²⁺-bound water may lead to dissociation of the HCO₃[−] molecule from the Co²⁺ ion, due to the charge–charge repulsion between the formed hydroxide ion and the HCO₃[−] molecule. Following the HCO₃[−] dissociation, the tetrahedral coordination is restored (Fig. 2c).

On the other hand, at 20 atm CO₂ pressure, Ni-CA II shows octahedral coordination comprising the bidentate HCO₃⁻ and a

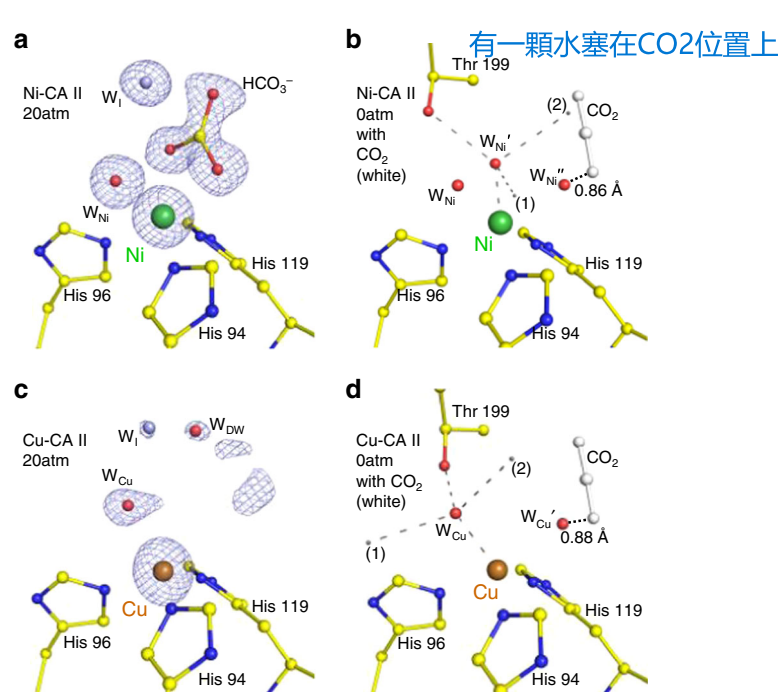


Fig. 5 Substrate/product binding in Ni- and Cu-CA II. The intermediate water (W_i) is colored in steel blue for clarity. The electron density ($2F_o - F_c$, blue) is contoured at 2.2σ . **a** At 20 atm of CO_2 pressure, Ni-CA II maintains octahedral coordination with HCO_3^- binding. **b** Compared to the W_{Zn} geometry in Zn-CA II (Fig. 3d), the nucleophilic attack geometry around W_{Ni} has steric hindrance on CO_2 molecule (adapted from Zn-CA II, 20 atm, white) and is distorted away. Distance between the position (2) and C atom of CO_2 is 1.55 Å. **c** Cu-CA II shows only disordered electron density in the $\text{CO}_2/\text{HCO}_3^-$ binding site. **d** The nucleophilic attack geometry around W_{Cu} has steric hindrance on CO_2 molecule (adapted from Zn-CA II, 20 atm) and is significantly distorted away. Distance between the position (2) and C atom of CO_2 is 2.93 Å.

water molecule in a similar manner to that of Co-CA II (Fig. 5a, Supplementary Fig. 1g-i). It is noted that one of the three bound water molecules experiences steric hindrance with the CO₂-binding configuration in Zn-CA II (Fig. 5b). Thus, it is likely that the CO₂ molecule entering the active site pushes away one of the Ni-bound water molecules, and then a nucleophilic attack occurs from one of the two remaining water molecules, forming HCO₃⁻. Unlike Co-CA II (Fig. 4a, c,d), the Ni-CA II intermediates obtained at different pH values (7.8 and 11.0) indicate that the HCO₃⁻ binding affinity is almost unresponsive to pH variation (Supplementary Fig. 3). The result suggests that the deprotonation of the Ni²⁺-bound water is insufficient to facilitate HCO₃⁻ dissociation in the stable octahedral coordination, and that the bound HCO₃⁻ is directly displaced by two incoming water molecules in Ni-CA II. Finally, in Cu-CA II, no clear electron density of CO₂ or HCO₃⁻ is visible (Fig. 5c). The faint and diffused electron density suggests that a CO₂ molecule encounters a severe steric hindrance from one of Cu²⁺-bound water molecules. Even if the CO₂ molecule adopts proper orientation as in Zn-CA II, the bound CO₂ position remains far too distant (3.9 Å) from the spare Cu²⁺-bound water molecule for any effective interaction. Moreover, the significantly distorted geometry negates the scope of any nucleophilic attack (Fig. 5d). The inefficient substrate binding and the unfavorable distorted geometry explain the complete enzymatic inactivity of Cu-CA II.

Electrostatic effects of metal ions on active-site water network.

Figure 6 shows the proton transfer pathway and the water

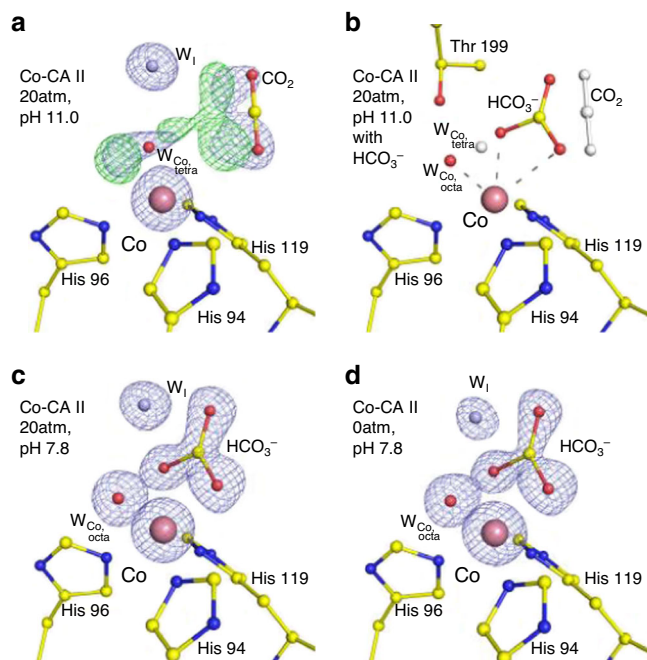


圖 4 Co-CA II 中的底物/產物結合。為了清楚起見，中間水 (W_1) 的顏色為銅藍色。電子密度 ($2F_o - F_c$ ，藍色) 和差異圖 ($F_o - F_c$ ，綠色) 分別在 2.2σ 和 7.0σ 處繪製。a, b 在 20 atm CO_2 壓力下，pH 11.0 下的 Co-CA II 顯示 CO_2 結合 (~50% 佔據率，白色) 與四面體配位的疊加和 HCO_3^- 結合 (~50% 佔據率) 與八面體配位率的疊加。c, d Co-CA II 在 pH 7.8 下顯示 HCO_3^- 的完全結合，即使在不添加 CO_2 的情況下也顯示出八面體配位。捕獲的 HCO_3^- 很可能是從環境空氣中吸收到晶體中的 CO_2 轉化而來的。

CO_2 分子取代了深水， W_{DW} 存在於活性位點內，顯示活性位點上的 CO_2 結合主要由不含金屬的蛋白質支架決定 (圖 3a)。天然全酶 Zn-CA II 顯示的 CO_2 結合幾乎與 apo-CA II 中的相同 (圖 3b、c)。 CO_2 分子距離 Zn 結合水 (W_{Zn}) 2.9 Å，其構型有利於親核攻擊 (圖 3d)。

在 20 atm CO_2 壓力下低溫冷卻的 Co-CA II (pH 11.0) 中，觀察到 CO_2 和 HCO_3^- 的雙重結合 (圖 4a)。 CO_2 結合後，四面體配位得以維持，但 HCO_3^- 結合後觀察到不尋常的八面體配位擴展 (圖 4b)。在變換後的八面體幾何結構中， HCO_3^- 分子以二齒模式與 Co^{2+} 離子以及額外的水分子結合。與 Zn-CA II 中的單齒結合模式相比，二齒 HCO_3^- 上的負電荷可以分佈在與 Co^{2+} 離子結合的兩個氧原子之間，從而使產物與金屬離子的結合更強 (補充圖 1a-f)。與 Zn-CA II (補充圖 2) 不同，在不同 pH 值 (7.8 和 11.0) 下獲得的 Co-CA II 中間體表明， HCO_3^- 分子與 Co^{2+} 離子牢固結合，並在較低 pH 值下完全佔據 (圖 4c, d)，但這種結合親和力隨著 pH 值的增加而減弱 (圖 4a)。結果表明，在催化循環期間，由於形成的氫氧根離子和 HCO_3^- 分子之間的電荷排斥， Co^{2+} 結合水的去質子化可能導致 HCO_3^- 分子從 Co^{2+} 離子解離。 HCO_3^- 解離後，四面體配位恢復 (圖 2c)。

另一方面，在 20 atm CO_2 壓力下，Ni-CA II 顯示出八面體配位，包括二齒 HCO_3^- 和

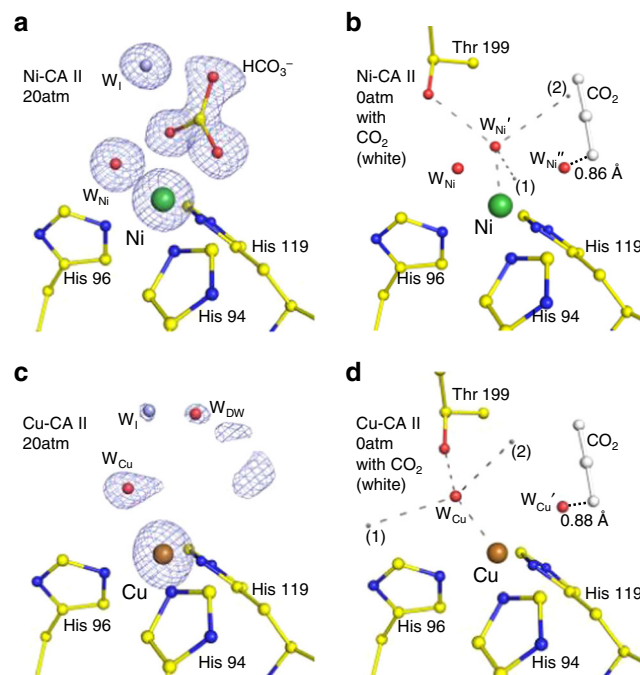
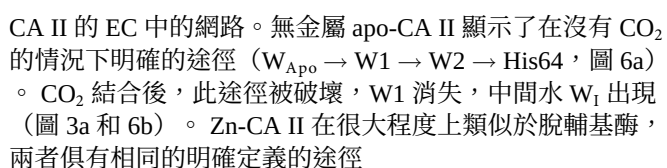


圖 5 Ni-和 Cu-CA II 中的底物/產物結合。為了澄清起見，中間水 (W_1) 的顏色為銅藍色。電子密度 ($2F_o - F_c$ ，藍色) 的輪廓為 2.2σ 。a 在 20 大氣壓的 CO_2 壓力下，Ni-CA II 與 HCO_3^- 結合保持八面體配位。b 與 Zn-CA II 中的 W_{Zn} 幾何結構相比 (圖 3d)， W_{Ni} 周圍的親核攻擊幾何結構對 CO_2 分子 (改編自 Zn-CA II，20 atm，白色) 具有空間位阻，並且被扭曲掉。位置 (2) 與 CO_2 的 C 原子之間的距離為 1.55 Å。c Cu-CA II 僅在 $\text{CO}_2/\text{HCO}_3^-$ 結合位點顯示無序電子密度。d W_{Cu} 周圍的親核攻擊幾何結構對 CO_2 分子 (改編自 Zn-CA II，20 atm) 具有空間位阻，並且明顯扭曲。位置 (2) 與 CO_2 的 C 原子之間的距離為 2.93 Å。

水分子以與 Co-CA II 類似的方式 (圖 5a，補充圖 1g-i)。值得注意的是，三個結合水分子之一與 Zn-CA II 中的 CO_2 結合構型存在空間位阻 (圖 5b)。因此，進入活性位點的 CO_2 分子很可能會推開其中一個與 Ni 結合的水分子，然後剩下的兩個水分子之一發生親核攻擊，形成 HCO_3^- 。與 Co-CA II 不同 (圖 4a、c、d)，在不同 pH 值 (7.8 和 11.0) 下獲得的 Ni-CA II 中間體顯示 HCO_3^- 結合親和力幾乎對 pH 變化沒有反應 (補充圖 3)。結果表明， Ni^{2+} 結合水的去質子化不足以促進穩定八面體配位中的 HCO_3^- 解離，並且結合的 HCO_3^- 直接被 Ni-CA II 中的兩個進入的水分子取代。最後，在 Cu-CA II 中，看不到明顯的 CO_2 或 HCO_3^- 電子密度 (圖 5c)。微弱且擴散的電子密度表明 CO_2 分子遇到了來自 Cu^{2+} 結合水分子之一的嚴重空間位阻。即使 CO_2 分子採用 Zn-CA II 中的正確方向，結合的 CO_2 位置仍然距離備用的 Cu^{2+} 結合水分子太遠 (3.9 Å)，無法進行任何有效的相互作用。此外，顯著扭曲的幾何形狀否定了任何親核攻擊的範圍 (圖 5d)。低效的底物結合和不利的扭曲幾何結構解釋了 Cu-CA II 的完全酶促失活。

金屬離子對活性位點水網絡的靜電效應。圖 6 顯示了質子轉移路徑和水



5

Fig. 6 Active site in CA II showing proton transfer pathway and EC water network ($W_{EC1} - W_{EC5}$). The electron density ($2F_o - F_c$) is contoured at 1.7σ except for EC waters at 1.5σ . The EC waters are colored in aqua marine and the intermediate waters (W_I and W_I') in steel blue for clarity. W_2' is an alternative position of W_2 . The possible proton transfer pathways in the metal-CA IIs are depicted as red arrows. All structures were obtained at pH 7.8 except for **e** at pH 11.0. **a, b** Apo-CA II shows well-ordered water arrangement (dotted red line) with His64 favored in outward conformation at 0 atm CO_2 pressure. Upon CO_2 binding, His64 moves inward and water molecules show highly dynamical motions, stabilizing W_I . **c, d** Zn-CA II shows His64 favored in inward conformation at 0 atm CO_2 pressure. Upon CO_2 binding, W_2 , W_3b , and W_{EC} waters show significantly different dynamics with His64 moving outward, and an additional intermediate water (W_I') is stabilized with the W_{EC} molecules. The motions of W_3b and W_{EC1} turn on the dynamic interplay between the proton transfer and EC water networks. **e, f** Co-CA II shows similar arrangement initially as in Zn-CA II. However, upon full HCO_3^- binding, the dynamical motions of EC waters are different and the intermediate water W_I' is less stabilized. Note that, in Co-CA II, proton transfer seems to occur while the product is still bound. **g, h** Ni-CA II initially shows altered water arrangements due to octahedral coordination. Upon HCO_3^- binding, significantly reduced water dynamical motions are recognized. **i, j** Cu-CA II shows unexpectedly similar dynamical motions of active site waters and His64 as in Zn-CA II.

modified dynamics of W_2 and His64, which are believed to be critical for efficient proton transfer (Fig. 6d). Additionally, in Zn-CA II, the significantly modified dynamics of the EC waters stabilizes another intermediate water W_I' , that in turn bridges W_I with the bulk solvent outside the protein, thereby facilitating the replenishment of W_{Zn} and W_1 during the catalytic cycle (Figs. 1b and 6d). These observations suggest that the Zn^{2+} ion produces a long-range (~ 10 Å) electrostatic field in which water structure and dynamics in the active site are fine-tuned to facilitate the proton transfer and the water/substrate/product exchange.

In the absence of CO_2 , Co-CA II forms the same proton transfer network as in Zn-CA II (Fig. 6e). Once CO_2 or HCO_3^- binds, W_1 disappears and W_I appears like what happens in Zn-CA II (Figs. 4a, c, and 6f). However, as the deprotonation of Co^{2+} -bound water should occur prior to the HCO_3^- dissociation, it is likely that the proton transfer occurs via the altered network ($W_{Co,octa} \rightarrow W_2 \rightarrow \text{His64}$) while the product is still bound to the Co^{2+} ion (Fig. 6f). In addition, Co-CA II shows modified dynamics of W_2 , His64, and EC waters as compared to Zn-CA II (Fig. 6f). Meanwhile, in Ni-CA II, octahedral coordination is stabilized throughout the entire catalytic cycle, and consequently, W_1 is absent due to its steric hindrance with one of the Ni-bound water molecules (Figs. 2d, 5a, and 6g). Based on the CO_2 binding configuration in Zn-CA II and the bidentate HCO_3^- binding observed in Ni-CA II, it is most likely that the substrate-to-product conversion occurs via the nucleophilic attack from W_{Ni}' to CO_2 (Fig. 5b). This result suggests that the proton transfer occurs possibly via the modified pathway $W_{Ni}' \rightarrow W_{Ni} \rightarrow W_2 \rightarrow \text{His64}$ (Fig. 6g). Also, unlike Zn-CA II and Co-CA II, W_2 in Ni-CA II shows significantly different dynamics and W_I' is destabilized, a plausible reflection of the altered electrostatic environment (Fig. 6h). Finally, Cu-CA II reveals that the possible proton transfer pathway ($W_{Cu} \rightarrow W_1 \rightarrow W_2 \rightarrow \text{His64}$) is well-defined (Fig. 6i) and W_2 and His64 dynamics is surprisingly similar to that in Zn-CA II (Fig. 6j). This result corroborates well with our conjecture that it is the lack of efficient substrate binding and unfavorable distorted geometry for the nucleophilic attack that are responsible for the complete inactivity of Cu-CA II.

Discussion

Our results provide advanced insights into the role of metal ions and the metal-protein relationship for the CA II catalytic mechanism. In the absence of metal ions, the protein scaffold provides a fundamental structural template necessary for the catalytic activity. The protein scaffold helps usher a substrate molecule from the outside bulk solvent into the active site through desolvating and positioning it at a configuration conducive for nucleophilic attack. The protein scaffold also provides well-ordered water networks in the vicinity of the active site, which can be utilized for proton transfer and substrate/product/water exchange. Metal ions then bring a key property for the CA II catalytic activity in generating hydroxyl ion at neutral pH and

retaining it at the active site. Beyond their primary Lewis acid property, metal ions are directly involved in the catalytic mechanism via their coordination geometry and long-range electrostatic effects. The most efficient native Zn-CA II preserves a tetrahedral coordination and fine-tunes the water network embedded within the protein scaffold (Fig. 1c). The tetrahedral coordination allows efficient conversion of substrate into product, and the long-range electrostatic field orchestrates the structure and dynamics of water network in the active site, imperative for the rapid product displacement and fast proton transfer. In comparison, semi-efficient Co-CA II shows similar catalytic behavior up to the product formation stage as in Zn-CA II, but the expansion of the metal coordination geometry from tetrahedron to octahedron during the catalytic cycle alters the product displacement and proton transfer process (Fig. 7). The significantly less efficient Ni-CA II maintains octahedral coordination and shows altered electrostatic effects, hampering efficient conversion from substrate to product, product displacement, and proton transfer (Fig. 8). Finally, completely inactive Cu-CA II suggests substantial steric hindrance encountered by the substrate in the active site and poor geometry for product conversion due to the trigonal bipyramidal coordination (Fig. 9).

In conclusion, we examined the role of various metal ions in carbonic anhydrase catalysis beyond their primary chemical property as a Lewis acid. We demonstrated that metal ions are directly involved in the enzymatic mechanism via their coordination geometry and long-range electrostatics to orchestrate intricate water dynamics. Our experimental results can be used as direct input for theoretical and computational studies on the role of metal ions, which we anticipate could open a new window to the study of metal-protein relationships, drug discovery targeting metalloenzymes, engineering of natural metalloenzymes, rational design of de novo metalloenzymes, and synthesis of supramolecular analogues to metalloenzymes.

Methods

Protein expression and purification. The native Zn-CA II was expressed in a recombinant strain of *Escherichia coli* [BL21 (DE3) pLysS] containing a plasmid encoding the CA II gene³⁵. Purification was carried out using affinity chromatography³⁶. Briefly, bacterial cells were enzymatically lysed with hen egg white lysozyme, and the lysate was placed onto an agarose resin coupled with *p*-(aminomethyl)-benzene-sulfonamide which binds CA II. The protein on the resin was eluted with 0.4 M sodium azide, in 100 mM Tris-HCl pH 7.0. The azide was removed by extensive buffer exchange against 10 mM Tris-HCl pH 8.0.

Apo-CA II (zinc free) was then prepared by incubating Zn-CA II in a zinc chelation buffer (100 mM pyridine 2,6-dicarboxylic acid, 25 mM MOPS pH 7.0) at 20 °C for 18 h. The resulting protein was then run through an affinity column with benzylsulfonamide resin to remove residual Zn-CA II. The chelating agent was then removed by buffer-exchange against 50 mM Tris-HCl pH 7.8¹⁸. The loss of zinc ion was examined using the esterase kinetic assay and further confirmed in the crystallographic structure. The enzyme activity was revived by an addition of 1 mM $ZnCl_2$.

Esterase kinetic assay. The CO_2/HCO_3^- conversion catalytic activity of CA II can be measured directly by stopped flow assays, monitoring labeled CO_2/HCO_3^-

圖 6 CA II 中的活性位點顯示質子轉移途徑和 EC 水網絡 ($W_{EC1} - W_{EC5}$)。電子密度 ($2F_o - F_c$) 的等高線為 1.7σ ，EC 水除外，為 1.5σ 。為了清晰起見，EC 水域採用水藍色，中間水域 (W_i 和 W_i') 採用銅藍色。W2' 是 W2 的替代位置。金屬-CA II 中可能的質子轉移途徑如紅色箭頭所示。所有結構均在 pH 7.8 下獲得，除了 e 是在 pH 11.0 下獲得的。a、b Apo-CA II 顯示良好有序的水排列（紅色虛線），在 0 atm CO_2 壓力下，His64 傾向於向外構象。CO₂ 結合後，His64 向內移動，水分子表現出高度動態的運動，從而穩定 W_i 。c、d Zn-CA II 顯示 His64 在 0 atm CO_2 壓力下傾向於向內構象。CO₂ 結合後，W2、W3b 和 W_{EC} 水錶現出顯著不同的動力學，His64 向外移動，並且額外的中間水 (W_i') 與 W_{EC} 分子一起穩定。W3b 和 W_{EC1} 的運動開啟了質子轉移和 EC 水網絡之間的動態交互作用。e、f Co-CA II 最初顯示出與 Zn-CA II 中類似的排列。然而，在完全 HCO_3^- 結合後，EC 水的動力學運動不同，且中間水 W_i' 較不穩定。請注意，在 Co-CA II 中，質子轉移似乎在產物仍處於結合狀態時發生。g、h Ni-CA II 最初顯示出由於八面體配位而改變的水排列。 HCO_3^- 結合後，水的動力學運動顯著減少。i、j Cu-CA II 顯示出與 Zn-CA II 中活性位點水和 His64 出乎意料相似的動態運動。

修改了 W2 和 His64 的動力學，被 于有效的 子移至 重要（6d）。此外，在 Zn-CA II 中，EC 水的動力學顯著改變，穩定了另一種中間水 W_i' ，進而將 W_i 與蛋白質外部的本體溶劑橋接起來，從而促進催化循環期間 W_{Zn} 和 $W1$ 的補充（圖 1b 和 6d）。這些觀察結果表明， Zn^{2+} 離子產生長程 (~10 Å) 靜電場，其中活性位點的水結構和動力學得到微調，以促進質子轉移和水/底物/產物交換。

在沒有 CO_2 的情況下，Co-CA II 形成與 Zn-CA II 相同的質子轉移網絡（圖 6e）。一旦 CO_2 或 HCO_3^- 結合， $W1$ 就會消失， W_i 就會出現，就像 Zn-CA II 中發生的情況一樣（圖 4a、c 和 6f）。然而，由於 Co^{2+} 結合水的去質子化應該在 HCO_3^- 解離之前發生，因此質子轉移很可能透過改變的網絡 ($W_{Co,octa} \rightarrow W2 \rightarrow His64$) 發生，而產物仍與 Co^{2+} 離子結合（圖 6f）。此外，與 Zn-CA II 相比，Co-CA II 顯示了 W2、His64 和 EC 水的動態變化（圖 6f）。同時，在 Ni-CA II 中，八面體配位在整個催化循環中保持穩定，因此，由於與 Ni 結合水分子之一的空間位阻， $W1$ 不存在（圖 2d、5a 和 6g）。基於 Zn-CA II 中的 CO_2 結合構型和 Ni-CA II 中觀察到的雙齒 HCO_3^- 結合，底物到產物的轉化很可能是透過從 W_{Ni} 到 CO_2 的親核攻擊發生的。5b)。這結果顯示質子轉移可能透過修飾的途徑 $W_{Ni}' \rightarrow W_{Ni} \rightarrow W2 \rightarrow His64$ 發生（圖 6g）。此外，與 Zn-CA II 和 Co-CA II 不同，Ni-CA II 中的 W2 顯示出顯著不同的動力學，且 W_i' 不穩定，這似乎是靜電環境改變的反映（圖 6h）。最後，Cu-CA II 揭示了可能的質子轉移途徑 ($W_{Cu} \rightarrow W1 \rightarrow W2 \rightarrow His64$) 是明確定義的（圖 6i），並且 W2 和 His64 動力學與 Zn-CA II 中的動力學驚人地相似（圖 6j）。這一結果很好地證實了我們的猜想，即缺乏有效的底物結合和不利於親核攻擊的扭曲幾何形狀是導致 Cu-CA II 完全失活的原因。

討論

我們的研究結果為 CA II 催化機制中金屬離子的作用以及金屬-蛋白質關係提供了深入的見解。在沒有金屬離子的情況下，蛋白質支架提供催化活性所需的基本結構模板。蛋白質支架通過去溶劑化並將其定位在有利於親核攻擊的構型，幫助將底物分子從外部本體溶劑引入活性位點。蛋白質支架還在活性位點附近提供有序的水網絡，可用於質子轉移和底物/產物/水交換。金屬離子為 CA II 在中性 pH 條件下生成氫氧根離子的催化活性帶來了關鍵特性，

將其保留在活性位點。除了其主要的路易斯酸性質外，金屬離子還透過其配位幾何形狀和長程靜電效應直接參與催化機制。最有效的天然 Zn-CA II 保留了四面體協調性並微調了嵌入蛋白質支架內的水網絡（圖 1c）。四面體配位允許底物有效轉化為產物，長程靜電場協調活性位點中水網絡的結構和動力學，這對於快速產物置換和快速質子轉移至關重要。相較之下，半高效 Co-CA II 在產物形成階段表現出與 Zn-CA II 相似的催化行為，但在催化循環過程中金屬配位幾何形狀從四面體擴展到八面體改變了產物置換和質子轉移過程（圖 7）。效率明顯較低的 Ni-CA II 保持了八面體配位，並顯示出改變的靜電效應，阻礙了從基材到產物的有效轉化、產物置換和質子轉移（圖 8）。最後，完全失活的 Cu-CA II 顯示底物在活性位點遇到了很大的空間位阻，並且由於三角雙錐配位而導致產物轉化的幾何形狀較差（圖 9）。

總之，我們研究了各種金屬離子在碳酸酐酶催化中的作用，超越了其作為路易斯酸的主要化學性質。我們證明金屬離子透過其配位幾何結構和長程靜電直接參與酵素機制，以協調複雜的水動力學。我們的實驗結果可以作為金屬離子作用的理論和計算研究的直接輸入，我們預計這可以為金屬-蛋白質關係的研究、針對金屬酶的藥物發現、天然金屬酶的工程、從頭金屬酶的合理設計以及金屬酶的超分子類似物的合成打開一扇新的窗口。

方法

蛋白質表現和純化。天然 Zn-CA II 在含有編碼 CA II 基因 35 的質粒的大腸桿菌重組菌株 [BL21 (DE3) pLysS] 中表達。使用親和層析法進行純化³⁶。簡而言之，用雞蛋白溶菌酶對細菌細胞進行酶裂解，然後將裂解物置於瓊脂糖樹脂上，該樹脂與 CA II 結合的對 (氨基甲基)-苯磺酰胺偶聯。樹脂上的蛋白質以 100 mM Tris-HCl pH 7.0 中的 0.4 M 疊氮化鈉洗脫。透過與 10 mM Tris-HCl pH 8.0 進行大量緩衝液交換來去除疊氮化物。

然後將 Zn-CA II 在鋅螯合緩衝液 (100 mM 吡啶 2,6-二甲酸, 25 mM MO PS pH 7.0) 中於 20 °C 孵育 18 小時來製備 Apo-CA II (不含鋅)。然後將所得蛋白質通過帶有苄基磺酰胺樹脂的親和柱以去除殘留的 Zn-CA II。然後透過與 50 mM Tris-HCl pH 7.818 進行緩衝液交換來去除螯合劑。使用酯酶動力學測定檢查鋅離子的損失，並在晶體結構中進一步證實。添加 1 mM $ZnCl_2$ 可恢復酵素活性。

酯酶動力學測定。CA II 的 CO_2/HCO_3^- 轉化催化活性可以透過停流測定直接測量，監測標記的 CO_2/HCO_3^-

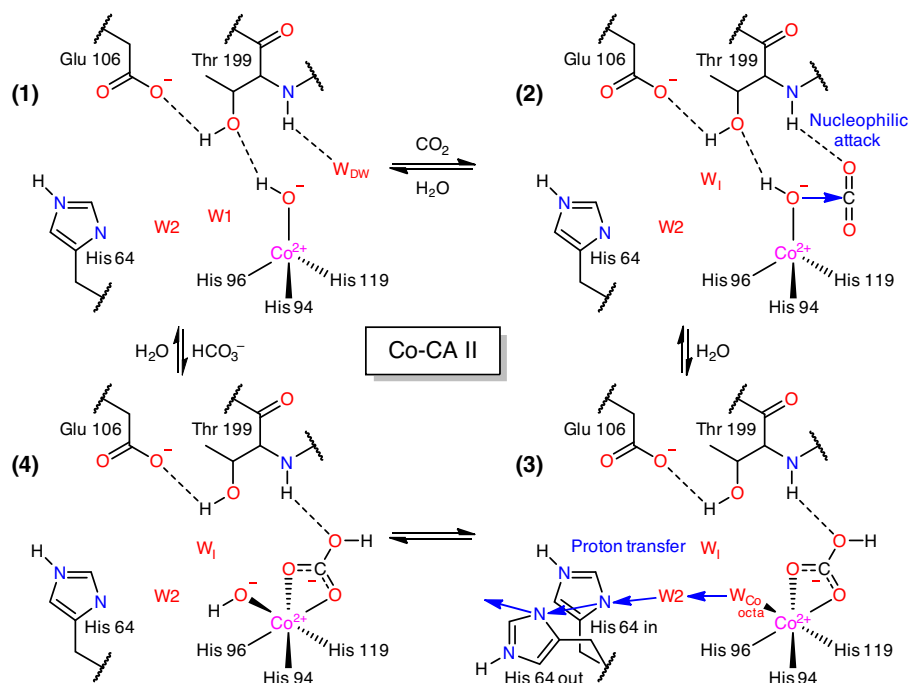


Fig. 7 Proposed catalytic mechanism of Co-CA II. In Co-CA II, CO_2 binding and the catalytic conversion of CO_2 to HCO_3^- occur in the same way as in the Zn-CA II with tetrahedral geometry. However, the HCO_3^- displacement and proton transfer process are significantly altered due to the coordination expansion to octahedral geometry during catalysis. This octahedral coordination allows bidentate binding mode of HCO_3^- and reorganization of negative charge of HCO_3^- toward Co^{2+} ion, allowing stronger HCO_3^- binding to metal ion. To dissociate the product, proton transfer first occurs via an altered pathway (possibly, $W_{\text{Co,octa}} \rightarrow W_2 \rightarrow \text{His64}$) and $W_{\text{Co,octa}}$ is converted into the hydroxyl ion. This negatively charged hydroxyl ion then pushes away the bound product, and the tetrahedral coordination is restored for the next catalytic cycle.

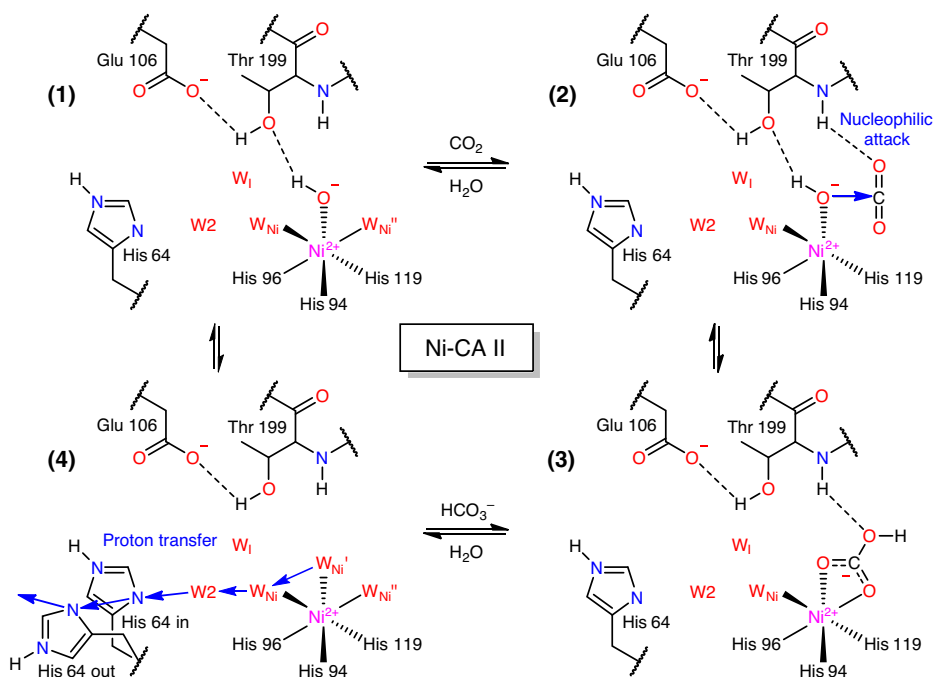


Fig. 8 Proposed catalytic mechanism of Ni-CA II. In Ni-CA II, octahedral coordination is maintained throughout the whole catalytic cycle. The significant consequence is that one of the three bound water molecules experiences steric hindrance with the CO_2 binding. In addition, the nucleophilic attack geometry is distorted (Fig. 5b), suggesting less efficient conversion into HCO_3^- . The formed HCO_3^- is strongly bound to Ni^{2+} ion in a bidentate mode as in the Co-CA II but is directly displaced by two inflowing water molecules. Finally, proton transfer occurs via an altered network (possibly, $W_{\text{Ni}}' \rightarrow W_{\text{Ni}} \rightarrow W_2 \rightarrow \text{His64}$) to restore the catalytic cycle.

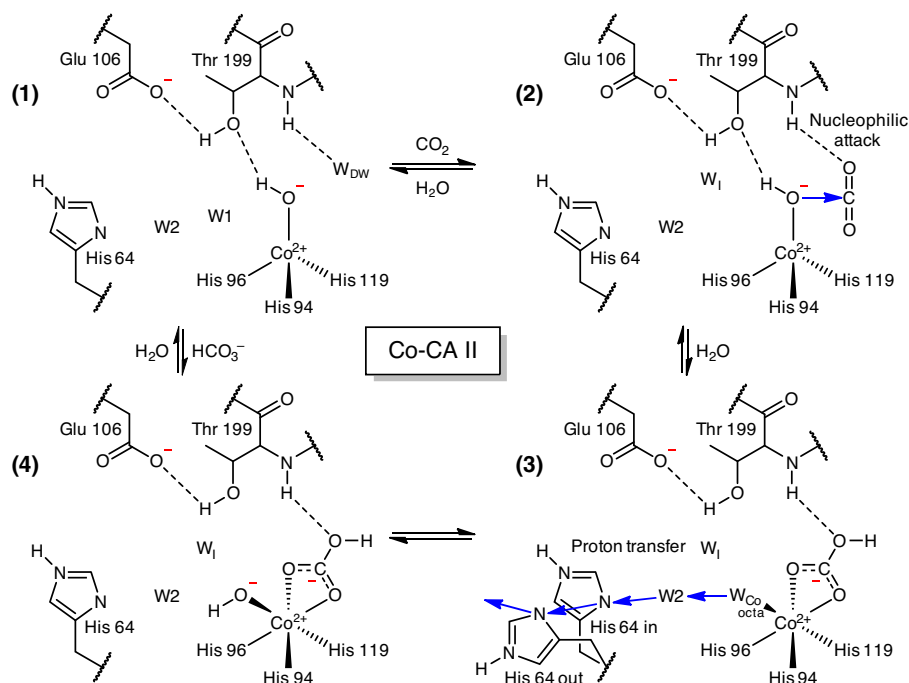


圖 7 Co-CA II 的建議催化機制。在 Co-CA II 中， CO_2 結合以及 CO_2 催化轉化為 HCO_3^- 的發生方式與四面體幾何形狀的 Zn-CA II 相同。然而，由於催化過程中配位擴展到八面體幾何形狀， HCO_3^- 位移和質子轉移過程發生了顯著變化。這種八面體配位允許 HCO_3^- 的雙齒結合模式以及 HCO_3^- 的負電荷向 Co^{2+} 離子重組，從而允許 HCO_3^- 與金屬離子更強的結合。為了解離產物，質子轉移首先透過改變的途徑發生（可能是 $\text{W}_{\text{Co,octa}} \rightarrow \text{W}_2 \rightarrow \text{His64}$ ），然後 $\text{W}_{\text{Co,octa}}$ 轉化為羧基離子。然後，帶負電荷的氫氧根離子會推開結合的產物，並恢復四面體配位以進行下一個催化循環。

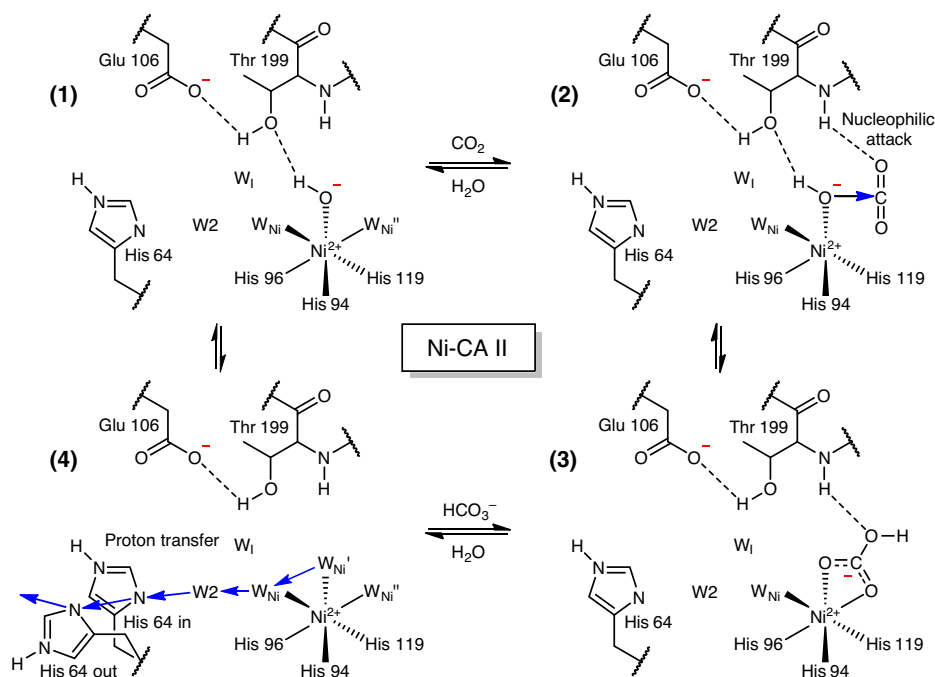


圖 8 Ni-CA II 的建議催化機制。在 Ni-CA II 中，八面體配位在整個催化循環中得以維持。重要的結果是三個結合水分子之一與 CO_2 結合經歷了空間位阻。此外，親核攻擊的幾何形狀被扭曲（圖 5b），顯示轉化為 HCO_3^- 的效率較低。形成的 HCO_3^- 以雙齒模式與 Co-CA II 中的 Ni^{2+} 離子牢固結合，但直接被兩個流入的水分子取代。最後，質子轉移是透過改變的網絡（可能是 $\text{W}_{\text{Ni}}' \rightarrow \text{W}_{\text{Ni}} \rightarrow \text{W}_2 \rightarrow \text{His64}$ ）發生，以恢復催化循環。

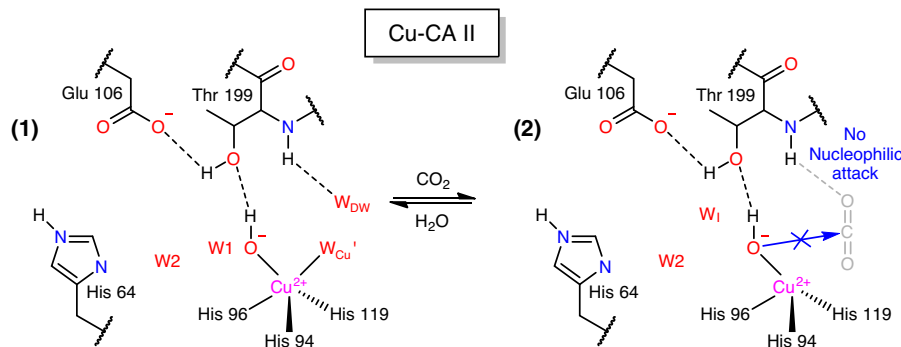


Fig. 9 Proposed catalytic mechanism of Cu-CA II. In Cu-CA II, one of the two bound water molecules in the trigonal bipyramid coordination experiences significant steric hindrance from CO₂ molecule, hindering adoption of proper configuration for nucleophilic attack. In addition, even if CO₂ binds to the active site temporarily, the nucleophilic attack geometry is too distant (3.9 Å) and significantly distorted (Fig. 5d).

conversion using mass spectroscopy, or indirectly by monitoring the innate esterase activity spectroscopically^{37,38}. In this study, the esterase activity assays were performed as a control to ensure zinc was fully chelated from recombinant CA II. The 4-nitrophenyl acetate (4-NPA) molecule is cleavable by CA II and thus used here as a colorimetric substrate. CA II cleaves the ester bond of 4-NPA generating 4-nitrophenol, which is spectroscopically absorbent at 348 nm in the ultraviolet-visible spectrum. Thus, the reaction can be monitored spectroscopically at 348 nm³⁹.

In a 96 deep-well plate, aliquots of 50 µL of 0.1 mg mL⁻¹ CA II in storage buffer were added to each well. To initiate the reaction, 200 µL of 0.8 mM 4-NPA dissolved in 3% acetone in water was added to the sample well. The well plate was then immediately inserted into the plate reader (Synergy HTX, BioTek, Winooski, WI, USA). Absorbance at 348 nm was recorded every 8 s for 10 min. The absorbance data of Apo- and Zn-CA II are plotted in Supplementary Fig. 4.

Crystallization and non-native metal substitution. Crystals of CA II were obtained using the hanging drop vapor diffusion method⁴⁰. A 10 µL drop of equal volumes of protein (5 µL) and the well-solution (5 µL) was equilibrated against 500 µL of the well-solution (1.3 M sodium citrate, 50 mM Tris-HCl pH 7.8) at RT (~20 °C)⁴¹. Crystals grew to an approximate ~30 × 100 × 200 µm³ in size in a few days. To prepare non-native metal substituted CA II, the apo-CA II crystals were transferred into soaking solutions of cobalt, nickel and copper salt (100 mM CoCl₂, 100 mM NiCl₂, 10 mM CuCl₂ along with 1.3 M sodium citrate, 50 mM Tris-HCl with pH 7.8). The crystals were incubated for 2–3 days to let the Co²⁺, Ni²⁺ and Cu²⁺ ions infuse into the active site⁴². The CA II crystals at pH 11.0 were obtained with 3-(cyclohexylamino)propanesulfonic acid buffer instead of using Tris-HCl.

Cryocooling under CO₂ pressure. Cryo-trapping the intermediate states of Zn-CA II was previously achieved by cryocooling CA II crystals under CO₂ pressure^{33,34}, leading to the capture of CO₂ in the active site of CA II⁴³. More recently, series of intermediate states have been tracked in CA II by controlling the internal CO₂ pressure levels^{25,44}. In this study, the CO₂ entrapment was carried out using a high-pressure cryo-cooler for X-ray crystallography (HPC-201, Advanced Design Consulting, USA). The apo-, Zn-, Co-, Ni-, and Cu-CA II crystals were first soaked in a cryo-solution containing 35% (v/v) glycerol supplemented to the soaking solution. The crystals were then coated with mineral oil to prevent dehydration, and loaded into the base of high-pressure tubes³³. The coated mineral oil worked as a CO₂ buffering medium as well, aiding in the absorption of CO₂ into the crystals⁴⁵. The crystals were pressurized at room temperature in the pressure tubes with CO₂ gas at 0 atm (no pressurization) and 20 atm. After a wait of about 5 min, the crystals were cryocooled in liquid nitrogen (77 K). Once the CO₂ bound crystals were fully cryocooled, the CO₂ gas pressure was withdrawn, and the crystal samples were stored in a liquid nitrogen dewar for subsequent X-ray data collection.

X-ray diffraction and data collection. Diffraction data were collected at Pohang Light Source II (wavelength of 0.9793 Å, beam size of 100 µm) under nitrogen cold stream (100 K). Data were collected using the oscillation method in intervals of 1° step on an ADSC Quantum 270 CCD detector (Area Detector Systems Corporation, USA) with a crystal-to-detector distance of 120 mm. A total of 360 images were collected on each of the CA II crystal data sets.

For each data set, a new fresh pressure-cryocooled crystal was used. The absorbed X-ray dose for a single data set was less than 5 × 10⁵ Gy, which is much less than the Henderson dose limit of 1.2 × 10⁷ Gy⁴⁶. Moreover, we have checked that X-ray radiation dose at least up to 10⁷ Gy does not induce apparent changes in the active site. The result confirms that the active site structures described in our study are unaffected by the X-ray radiation. Indexing, integration, and scaling were performed by using HKL2000⁴⁷. The data processing statistics are given in Supplementary Table 1.

Structure determination and model refinement. The CA II structures were determined using the CCP4 program suite⁴⁸. Prior to refinement, a random 5% of the data were flagged for *R*_{free} analysis. The previously reported crystal structures (PDB codes of 5DSR and 5YUK for apo- and metal substituted CA II) were used as the initial phasing models^{25,49}. The maximum likelihood refinement (MLH) was carried out using REFMAC5⁵⁰. The refined structures were manually checked using the molecular graphics program COOT⁵¹. Iterations of MLH were carried out with anisotropic B factor.

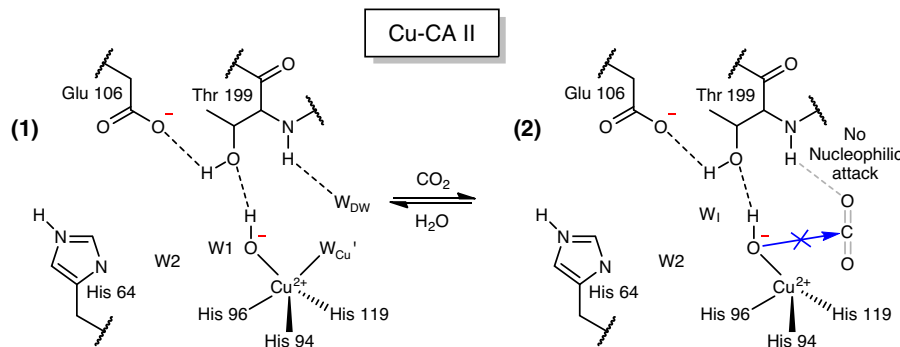
On completion of the structural refinements as described above, systematic refinements were further carried out to accurately determine the partial occupancies of the His 64 in and the His 64 out configurations. A total of 99 structures were prepared for each of the CA II structures, in which the occupancies of the His 64 in and the His 64 out configurations were changed in incremental steps of 1% (i.e., the first structure with 1% in and 99% out, the second structure with 2% in and 98% out, ..., the 99th structure with 99% in and 1% out). MLH refinements were carried out in parallel for all the 99 structures. After MLH refinements, the overall *R*-factor as a function of partial occupancy of the His 64 in configuration was obtained, and it was fitted into a quadratic function (Supplementary Fig. 5). The partial occupancy values of the His 64 configurations were determined where the overall *R*-factor is minimized. Details on the final refinement statistics are given in Supplementary Table 1. All structural figures were rendered with PyMol (Schrödinger, LLC).

Structural analysis of the bound water molecules. To compare the bound water molecules in the active site and the EC, we carefully refined water molecules based on the PDB and COOT validation checks and the electron density maps (cutoff level of 1σ in 2F_o–F_c electron density map). We have tested the consistency and reproducibility of the bound water molecules in the active site and the EC of the CA II structures. Since most of these waters exist transiently, it was allowed that they can be located closer than the normal stably bound water molecules. In this regard, water molecules closely located near the active site and EC regions were not excluded in the final coordinates. The important bound water molecules addressed in the main paper are listed in Supplementary Table 2. The distance information between CO₂, HCO₃⁻, Thr199, and important water molecules is listed in Supplementary Table 3.

Reporting summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

The atomic coordinates and structure factors have been deposited in the Protein Data Bank (<http://www.pdb.org/>) as [PDB code 6LUU [<https://doi.org/10.2210/pdb6luu/pdb>] (0 atm CO₂ pressure, pH 7.8), 6LUV [<https://doi.org/10.2210/pdb6luv/pdb>] (20 atm, pH 7.8)] for apo-CA II, [6LUW [<https://doi.org/10.2210/pdb6luw/pdb>] (0 atm, pH 7.8), 6LUX [<https://doi.org/10.2210/pdb6lux/pdb>] (20 atm, pH 7.8), 6LUY [<https://doi.org/10.2210/pdb6luy/pdb>] (0 atm, pH 11.0), 6LUZ [<https://doi.org/10.2210/pdb6luz/pdb>] (20 atm, pH 11.0)] for Zn-CA II, [6LV1 [<https://doi.org/10.2210/pdb6lv1/pdb>] (0 atm, pH 7.8), 6LV2 [<https://doi.org/10.2210/pdb6lv2/pdb>] (20 atm, pH 7.8), 6LV3 [<https://doi.org/10.2210/pdb6lv3/pdb>] (0 atm, pH 11.0), 6LV4 [<https://doi.org/10.2210/pdb6lv4/pdb>] (20 atm, pH 11.0)] for Co-CA II, [6LV5 [<https://doi.org/10.2210/pdb6lv5/pdb>] (0 atm, pH 7.8), 6LV6 [<https://doi.org/10.2210/pdb6lv6/pdb>] (20 atm, pH 7.8), 6LV7 [<https://doi.org/10.2210/pdb6lv7/pdb>] (0 atm, pH 11.0), 6LV8 [<https://doi.org/10.2210/pdb6lv8/pdb>] (20 atm, pH 11.0)] for Ni-CA II, and [6LV9 [<https://doi.org/10.2210/pdb6lv9/pdb>] (0 atm, pH 7.8), 6LVA [<https://doi.org/10.2210/pdb6lva/pdb>] (20 atm, pH 7.8)] for Cu-CA II. Two earlier structures [5DSR [<https://doi.org/10.2210/pdb5dsr/pdb>] and 5YUK



9 Cu-CA II 的拟 催化机制。在 Cu-CA II 中，三角雙錐配位中的兩個結合水分子之一受到 CO₂ 分子的顯著空間位阻，阻礙了親核攻擊採用適當的構型。此外，即使 CO₂ 暫時結合到活性位點，親核攻擊幾何結構也太遠 (3.9 Å) 並且顯著扭曲 (圖 5d)。

使用質譜法進行轉化，或透過光譜監測先天酯酶活性間接轉化^{37,38}。在本研究中，酯酶活性測定作為對照，以確保鈣與重組 CA II 完全整合。乙酸 4-硝基苯酯 (4-NPA) 分子可被 CA II 裂解，因此在此用作比色底物。CA II 裂解 4-NPA 的酯鍵，生成 4-硝基苯酚，在紫外可見光光譜中的 348 nm 處具有光譜吸收性。因此，可以在 348 nm 處透過光譜監測反應³⁹。

在 96 深孔板中，將 50 μ L 0.1 mg mL⁻¹ CA II 的儲存緩衝液等分樣本加入每個孔中。為了引發反應，將 200 μ L 溶解在 3% 丙酮水溶液中的 0.8 mM 4-NPA 加入樣品孔中。然後立即將孔板插入讀板器 (Synergy HTX, BioTek, Winooski, VT, USA)。每 8 秒記錄一次 348 nm 處的吸光度，持續 10 分鐘。Apo- 和 Zn-CA II 的吸光度數據繪製在補充圖 4 中。

結晶和非天然金屬取代。CA II 晶體是使用懸滴蒸氣擴散法獲得的。10 μ L 滴等體 的蛋白 (5 μ L) 和孔溶液 (5 μ L) 在 RT (~20 °C) 下與 500 μ L 孔溶液 (1.3 M 檸檬酸，50 mM Tris-HCl pH 7.8) 平衡⁴¹。晶體在幾天內生長到大約 30 $\times$ 100 $\times$ 200 μ m³ 的大小。為了製備非天然金屬取代的 CA II，apo-CA II 晶體 移到、和銅 的浸泡溶液中 (100 mM CoCl₂、100 mM NiCl₂、10 mM CuCl₂ 以及 1.3 M 檸檬酸、50 mM Tris-HCl, pH 7.8)。將晶體孵育 2-3 天，讓 Co²⁺、Ni²⁺ 和 Cu²⁺ 離子注入活性位點⁴²。使用 3-(環己基)丙磺酸緩衝液取代 Tris-HCl 以獲得 pH 11.0 的 CA II 晶體。

在 CO₂ 壓力下低溫冷卻。冷凍捕獲 Zn-CA II 的中間態之前是透過在 CO₂ 壓力下冷凍 CA II 晶體來實現的^{33,34}，從而在 CA II₄₃ 的活性位點捕獲 CO₂。最近，透過控制內部 CO₂ 壓力水平，在 CA II 中追蹤了一系列中間狀態^{25,44}。在這項研究中，CO₂ 的捕獲是使用 X 射線晶體學高壓低溫冷卻器 (HPC-201, Advanced Design Consulting, 美國) 進行的。首先將 apo-、Zn-、Co-、Ni- 和 Cu-CA II 晶體浸泡在含有 35% (v/v) 甘油的冷凍溶液中，並補充浸泡溶液。然後將晶體塗上礦物油以防止脫水，並將其裝入高壓管的底座中³³。塗覆的礦物油也作為 CO₂ 緩衝介質，有助於將 CO₂ 吸收到晶體中⁴⁵。晶體在室溫下在壓力管中以 CO₂ 氣體在 0 atm (無加壓) 和 20 atm 下加壓。等待約 5 分鐘後，將晶體在液態氮 (77 K) 中低溫冷卻。一旦 CO₂ 結合晶體完全低溫冷卻，即撤消 CO₂ 氣壓，並將晶體樣品儲存在液態氮杜瓦瓶中以供後續 X 射線資料收集。

X 射線衍射和數據收集。衍射數據是在氮氣冷流 (100 K) 下在浦項光源 II (波長為 0.9793 Å，光束尺寸為 100 μ m) 收集的。使用振盪方法在 ADSC Quantum 270 CCD 偵測器 (Area Detector Systems Corporation, 美國) 上以 1° 步長間隔收集數據，晶體到偵測器的距離為 120 mm。每個 CA II 晶體資料集總共收集了 360 張影像。

對於每個數據集，都使用新的新鮮壓力冷凍晶體。單一資料集吸收的 X 射線劑量小於 5 $\times$ 10⁵ Gy，遠低於亨德森劑量限值 1.2 $\times$ 10⁷ Gy⁴⁶。此外，我們已經檢查，至少高達 107 Gy 的 X 射線輻射劑量不會引起活性部位的明顯變化。結果證實我們研究中所描述的活性位點結構不受 X 射線輻射的影響。使用 HKL2000⁴⁷ 執行索引、整合和縮放。數據處理統計請見附表 1。

結構確定和模型細化。CA II 結構是使用 CCP4 程式套件⁴⁸ 確定的。在細化之前，隨機 5% 的資料被標記用於 R_{free} 分析。先前報導的晶體結構 (脫輔基和金屬取代的 CA II 的 PDB 代碼為 5DSR 和 5YUK) 被用作初始定相模型^{25,49}。使用 REFMAC5⁵⁰ 進行最大似然細化 (MLH)。使用分子圖形程式 COOT⁵¹ 手動檢查精煉的結構。使用各向異性 B 因子重複 MLH。

完成上述結構細化後，進一步進行系統細化，以準確確定 His 64 in 和 His 64 out 配置的部分佔用情況。為每個 CA II 結構總共準備了 99 個結構，其中 His 64 輸入和 His 64 輸出配置的佔用率以 1% 的增量步驟改變 (即，第一個結構具有 1% 輸入和 99% 輸出，第二個結構具有 2% 輸入和 98% 輸出，...，第 99 個結構具有 99% 輸入和 1% 輸出)。MLH 改進對所有 99 個結構同時進行。經過 MLH 細化後，獲得了整體 R 因子作為配置中 His 64 部分佔據的函數，並將其擬合成二次函數 (補充圖 5)。His 64 配置的部分佔用率值是在總體 R 因子最小化的情況下確定的。最終細化統計數據的詳細資訊在補充表 1 中給出。所有結構圖均使用 PyMol (Schrödinger, LLC) 繪製。

結合水分子的結構分析。為了比較活性位點和 EC 中的結合水分子，我們根據 PDB 和 COOT 驗證檢查以及電子密度圖 (2F_o - F_c 電子密度圖中 1 σ 的截止水平) 仔細細化了水分子。我們仔細測試了活性位點和 EC 中結合水分子的一致性和再現性。CA II 結構的活性位點和 EC 中有幾個位置緊密的水分子。由於大多數這些水是短暫存在的，因此它們可以比正常穩定結合的水分子更靠近。在這方面，最終座標中並未排除靠近活性位點和 EC 區域的水分子。主論文中提到的重要結合水分子列於補充表 2。CO₂、HCO₃⁻、Thr199 和重要水分子之間的距離資訊列於補充表 3。

報告摘要。有關研究設計的更多信息，請參閱本文連結的《自然研究報告摘要》。

數據可用性

原子座標和結構因子已儲存在蛋白質資料庫 (<http://www.pdb.org/>) 中，作為 [PDB 代碼 6LUU [<https://doi.org/10.2210/pdb6luu/pdb>] (0 atm CO₂ 壓力, pH 7.8), 6LUV [<https://doi.org/10.2210/pdb6luv/pdb>] (20 atm CO₂ 壓力, pH 7.8), 6LUW [<https://doi.org/10.2210/pdb6luw/pdb>] (0 atm, pH 7.8), 6LUX [<https://doi.org/10.2210/pdb6lux/pdb>] (20 atm, pH 7.8), 6LUY [<https://doi.org/10.2210/pdb6luy/pdb>] (0 atm, pH 11.0), 6LUZ [<https://doi.org/10.2210/pdb6luz/pdb>] (20 atm, pH 11.0)] 用於 Zn-CA II, [6LV1 [<https://doi.org/10.2210/pdb6lv1/pdb>] (0 atm, pH 7.8), 6LV2 [<https://doi.org/10.2210/pdb6lv2/pdb>] (20 atm, pH 7.8), 6LV3 [<https://doi.org/10.2210/pdb6lv3/pdb>] (0 atm, pH 11.0), 6LV4 [<https://doi.org/10.2210/pdb6lv4/pdb>] (20 atm, pH 11.0)] 用於 Co-CA II, [6LV5 [<https://doi.org/10.2210/pdb6lv5/pdb>] (0 atm, pH 7.8), 6LV6 [<https://doi.org/10.2210/pdb6lv6/pdb>] (20 atm, pH 7.8), 6LV7 [<https://doi.org/10.2210/pdb6lv7/pdb>] (0 atm, pH 11.0), 6LV8 [<https://doi.org/10.2210/pdb6lv8/pdb>] (20 atm, pH 11.0), 6LV9 [<https://doi.org/10.2210/pdb6lv9/pdb>] (0 atm, pH 7.8), 6LVA [<https://doi.org/10.2210/pdb6lva/pdb>] (20 atm, pH 7.8)] Cu-CA II。兩個早期結構 [5DSR [<https://doi.org/10.2210/pdb5dsr/pdb>] 和 5YUK

[<https://doi.org/10.2210/pdb5yuk/pdb>] were used for structure determination. Source data are provided with this paper.

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Author contributions

C.U.K. conceived the research, J.K.K., C.L., S.W.L., J.T.A. ran the experiments, J.K.K. and C.U.K. analysed the data. J.K.K., A.A., R.M., C.-M.G., and C.U.K. wrote the paper. All authors contributed to the overall scientific interpretation and edited the paper.

Competing interests

The authors declare no competing interests.

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Additional information

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Correspondence and requests for materials should be addressed to C.U.K.

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附加資訊

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